

Self-Explainable AI in Medical Imaging: A Comprehensive Literature Review on CLIP, Knowledge Graphs, and Zero-Shot Localization

Abstract

The deployment of Deep Learning (DL) in high-stakes medical environments is currently hindered by the “black box” nature of state-of-the-art models. While Vision-Language Models (VLMs) like CLIP have revolutionized zero-shot recognition, their application in clinical diagnostics requires rigorous interpretability and grounding in established medical knowledge. This review critically examines the transition from post-hoc explainability to Self-Explainable AI (S-XAI), analyzing methodologies that integrate Knowledge Graphs (KGs) and medical ontologies directly into model architectures. I provide an exhaustive analysis of recent frameworks such as MedKLIP, ConceptCLIP, and R2GenKG, evaluating their efficacy in zero-shot localization and report generation. Furthermore, I survey advanced evaluation metrics including TCAR and Diverse Sensitivity (DS), identifying significant gaps in current research regarding the alignment of visual features with semantic clinical concepts. This review establishes the theoretical foundation for a novel framework integrating Hybrid KGs with CLIP to enhance faithfulness and clinical utility.

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1 Introduction

The integration of Artificial Intelligence (AI) into medical imaging has catalyzed a paradigm shift in diagnostic radiology, transitioning from purely manual interpretation to advanced Computer-Aided Diagnosis (CAD) systems capable of detecting pathologies with near-human or super-human precision. Deep learning architectures, particularly Convolutional Neural Networks (CNNs) and increasingly Vision Transformers (ViTs), have demonstrated remarkable efficacy in tasks ranging from pneumonia detection in chest radiographs to malignancy classification in mammography [1]. However, the deployment of these models in high-stakes clinical environments remains severely constrained by the “black box” problem—the inherent opacity of the internal decision-making processes of complex non-linear models.

In clinical practice, a correct diagnosis is insufficient; it must be accompanied by a transparent, verifiable justification to ensure trust, facilitate accountability, and meet stringent regulatory standards such as the GDPR’s “right to explanation” [2]. This necessity has given rise to the critical field of Explainable AI (XAI). Historically, XAI in medicine has relied on post-hoc interpretations—techniques designed to explain a model’s decision after the fact without altering its internal structure. While methods like Grad-CAM and SHAP have become ubiquitous, they suffer from significant limitations regarding clinical fidelity, often highlighting irrelevant regions or failing to capture fine-grained pathological features.

Consequently, there is a growing consensus within the research community that medical AI requires a fundamental shift toward *Self-Explainable AI* (S-XAI). Unlike post-hoc approaches, S-XAI models are designed to be inherently interpretable, integrating explainability constraints directly into the model architecture and training process. This ensures that the explanations generated are not merely plausible rationalizations but are faithful representations of the model’s actual reasoning process [2,3].

A pivotal enabler for the next generation of S-XAI is the convergence of Vision-Language Models (VLMs), specifically those based on Contrastive Language-Image Pretraining (CLIP), with structured knowledge representations such as Knowledge Graphs (KGs) and medical ontologies. CLIP-based models have demonstrated unprecedented zero-shot capabilities, enabling the recognition and localization of unseen medical conditions by aligning visual features with semantic text embeddings in a shared latent space [1,4]. However, standard VLMs often learn statistical correlations from noisy web data rather than causal medical logic. By integrating Knowledge Graphs—structured repositories of medical facts (e.g., “Pleural Effusion is located in the Pleural Space”)—researchers aim to ground these powerful visual models in verifiable medical reasoning. Frameworks such as MedKLIP [5], RadGraph [6], and ConceptCLIP [7] represent the frontier of this integration, attempting to combine the generalization power of large-scale pretraining with the precision of symbolic medical knowledge.

2 Explainable AI in Medical Imaging

The landscape of XAI in medical imaging is broadly categorized into post-hoc explainability and self-explainability. Understanding the limitations of the former is essential to appreciating the necessity of the latter, particularly in the context of safety-critical diagnostics.

2.1 Post-hoc Explainability (The Traditional Approach)

Post-hoc methods attempt to interpret a trained model’s behavior without modifying its internal architecture. These techniques are widely used due to their model-agnostic nature but suffer from significant drawbacks regarding clinical fidelity and resolution.

2.1.1 Attribution-Based Methods

Attribution-based methods, also known as saliency or feature importance methods, aim to visualize which regions of an input image contributed most to a specific prediction. **Gradient-weighted Class Activation Mapping (Grad-CAM)** remains the most prominent technique in this category. It computes the gradients of the target class score with respect to the feature maps of the last convolutional layer to generate a coarse localization map highlighting relevant regions [2]. While Grad-CAM provides a quick visual check, its resolution is fundamentally limited by the size of the final feature map (often 7×7 or 14×14 pixels), making it notoriously unreliable for identifying fine-grained medical details such as small lung nodules or microcalcifications. The coarse heatmaps often “bleed” into surrounding healthy tissue, creating a false sense of localization accuracy. **Layer-wise Relevance Propagation (LRP)** addresses some of these resolution issues by backpropagating the prediction score (relevance) from the output layer to the input pixels using specific decomposition rules. Unlike gradient-based methods, LRP provides pixel-level granularity, which is theoretically more suitable for medical imaging [2]. However, empirical studies suggest that LRP heatmaps can be noisy and difficult for radiologists to interpret without significant filtering.

2.1.2 Perturbation-Based Methods

Perturbation-based methods probe the model by systematically altering the input and observing changes in the output confidence. **LIME (Local Interpretable Model-agnostic Explanations)** approximates the complex non-linear decision boundary of a black-box model locally with a simple, interpretable linear model. It achieves this by perturbing the input (e.g., masking superpixels) and training a surrogate model to mimic the black box’s behavior in that local neighborhood. A critical flaw of LIME in medical imaging is the generation of out-of-distribution samples; simply graying out parts of an X-ray or MRI can result in biologically implausible images, potentially leading the model to behave erratically and producing misleading explanations. **SHAP (SHapley Additive exPlanations)** offers a more theoretically grounded approach

based on cooperative game theory [8]. It assigns an importance value (Shapley value) to each feature, representing its marginal contribution to the prediction across all possible feature coalitions. While SHAP provides a unified measure of feature importance, it is computationally expensive, and like other post-hoc methods, it explains *correlations* used by the model but does not inherently reveal the *causal* or *semantic* reasoning behind a feature’s importance.

2.2 Self-Explainable AI (The Proposed Direction)

Self-Explainable AI (S-XAI) represents a paradigm shift from interpreting black boxes to designing “glass boxes.” S-XAI models embed interpretability constraints directly into the learning process, ensuring that the decision pathway is transparent and human-intelligible by design [2].

2.2.1 Input Explainability

Input explainability focuses on transforming raw data into meaningful, structured representations before processing. **Structured Features and Domain Knowledge:** Rather than feeding raw pixel arrays into dense networks, S-XAI approaches utilize explainable feature engineering. For instance, *Self-Interpretable Multiple Instance Learning (SI-MIL)* integrates deep convolutional features with explicit morphometric and spatial descriptors [9]. This ensures that the input features themselves carry defined clinical semantics, such as “nuclear size” or “tissue texture,” making the downstream decision process more traceable.

2.2.2 Model Explainability

This category encompasses architectures that enforce interpretable reasoning steps within the network itself. **Attention Mechanisms:** While standard attention is often treated as post-hoc visualization, *Structure-Guided Attention* architectures explicitly design attention modules to serve as explanation generators. The **SCOUTER** model [3], for example, utilizes slot attention to provide positive or negative evidence for specific categories without relying on a final linear classifier. This offers inherent “evidence” for a diagnosis, identifying exactly which visual slots contributed to the decision. **Prototype-Based Learning:** These models classify new images by comparing them to a learned set of “prototype” cases (exemplars) stored in the model’s weights. **ProtoAL** integrates this into an active learning framework, demonstrating that prototype-based reasoning can achieve high performance with fewer labels while providing intuitive case-based explanations (e.g., “This looks like Pneumonia because it is visually similar to Case A”) [10].

3 Knowledge-Guided & Concept-Based Learning

To bridge the semantic gap between low-level pixel features and high-level clinical diagnoses, modern S-XAI frameworks increasingly rely on explicit knowledge representations and concept-

based learning.

3.1 Structured Knowledge Representation

The integration of structured medical knowledge is a defining characteristic of SOTA S-XAI systems. This knowledge is typically encoded in Knowledge Graphs (KGs) and ontologies.

3.1.1 Knowledge Graphs (KGs)

Knowledge Graphs represent entities (nodes) and their relationships (edges).

- **Prior Knowledge Graphs:** These are constructed from expert-curated databases and provide a rigid, reliable backbone of medical facts. For example, a graph might encode the relationship Viral Pneumonia (Disease) $\xrightarrow{\text{causes}}$ Ground Glass Opacity (Finding). The **Unified Medical Language System (UMLS)** serves as the gold standard source for such graphs. Prior KGs ensure clinical validity and consistency but may struggle to adapt to novel patterns or subtle correlations found in new datasets. [5].
- **Data-Driven Knowledge Graphs:** Generated automatically by mining statistical correlations from large-scale medical datasets. While flexible, they may learn spurious correlations (e.g., associating “tubes” with “pneumothorax” due to treatment protocols rather than disease pathology). [6]
- **Hybrid Knowledge Graphs:** The most advanced systems, such as **R2GenKG**, employ hybrid graphs to leverage the strengths of both approaches. **R2GenKG** constructs a large-scale multi-modal medical knowledge graph (termed **M3KG**) using GPT-4o to extract entities and relations from ground truth reports and align them with disease-aware visual tokens. This construction involves three stages:
 - (1) using GPT-4o to generate training data for entity/relation models,
 - (2) building preliminary triplets
 - (3) extracting disease-aware visual patches to form the multi-modal graph.This allows the model to reason with both the reliability of expert knowledge and the nuance of visual patterns. [11].

3.1.2 Medical Ontologies

Ontologies provide the standardized vocabulary and hierarchical structure for KGs.

- **SNOMED CT and UMLS:** These ontologies underpin the semantic alignment in models like ConceptCLIP and MedKLIP. They organize medical concepts into rigorous hierarchies (e.g., Lung is a Body Part; Pneumonia is a Disease). Using standardized ontologies ensures that AI-generated explanations are precise, unambiguous, and interoperable with Electronic Health Records (EHRs). [12]

- **RadLex:** Specifically designed for radiology, RadLex is used in frameworks like Rad-Graph to standardize the extraction of entities (Anatomy, Observation) and relations (Modify, Located At, Suggestive Of) from radiology reports. This domain-specific ontology is crucial for capturing the nuances of radiological language, such as uncertainty ("cannot exclude") and negation ("no evidence of"). [6, 13]

3.2 Concept-Based Learning Mechanisms

Concept-based learning forces deep learning models to reason in terms of human-understandable concepts (e.g., "irregular border," "spiculated mass") rather than abstract feature vectors, thereby aligning the model's latent space with human cognition.

3.2.1 Supervised Learning

- **Concept Bottleneck Models (CBMs):** CBMs are a seminal architecture in S-XAI that split the classification process into two distinct steps: $x \rightarrow c$ (mapping input to concepts) and $c \rightarrow y$ (mapping concepts to prediction) [14]. A "concept bottleneck layer" is explicitly trained to predict the presence of specific attributes. This architecture allows clinicians to inspect the intermediate concepts and even *intervene* (edit) them if the model makes a mistake, enhancing trust and accuracy [15].
- **Concept Whitening:** This technique aligns the latent space axes of a neural network with known concepts, ensuring that specific dimensions in the feature space correspond to specific semantic concepts. This effectively "disentangles" the latent space, making individual neurons interpretable. [16]

3.2.2 Unsupervised and Automatic Discovery

- **Concept Activation Regions (CAR):** Addressing the limitation that concepts may not be linearly separable in the latent space (a key assumption of standard Concept Activation Vectors or CAVs [17]), CARs use kernel-based Support Vector Classifiers (SVCs) to define *regions* in the latent space where a concept resides [18]. This allows for the representation of complex, non-linear medical concepts (e.g., heterogeneous tissue textures).
- **ACE (Automated Concept-based Explanation):** The ACE algorithm automatically discovers visual concepts by clustering image segments and assessing their importance for prediction. It eliminates the need for expensive pixel-level concept annotations, which are rare in medical datasets.

3.2.3 Generative Concept Learning

- **CEIR (Concept-based Explainable Image Representation):** CEIR represents a significant leap by leveraging Large Language Models (LLMs) like GPT-4 to generate a pool

of interpretable concepts and CLIP to align them with images [19]. The model projects images into a “concept vector space” defined by these generated concepts, enabling open-world concept mining without expensive manual annotation.

- **LaBo (Language-Guided Bottlenecks):** Similar to CEIR, LaBo uses LLMs to define the concept space, creating a bottleneck layer that is semantically rich without requiring manual annotation. [20]

4 Vision-Language Models & Zero-Shot Localization

The advent of Vision-Language Models (VLMs), particularly those built on the CLIP architecture, has revolutionized zero-shot learning (ZSL) in medical imaging. By learning a shared embedding space for images and text, these models can identify and localize pathologies they have never seen during training, provided they can be described in text.

4.1 CLIP Architecture & Mechanism

CLIP (Contrastive Language-Image Pretraining) consists of two parallel encoders: a Vision Transformer (ViT) or ResNet for images, and a Transformer for text. The core mechanism is Contrastive Learning, specifically the InfoNCE loss function. This loss maximizes the cosine similarity between matched image-text pairs (positive pairs) while minimizing it for mismatched pairs (negative pairs) in a batch.

$$\mathcal{L}_{InfoNCE} = -\log \frac{\exp(\text{sim}(I, T)/\tau)}{\sum_j \exp(\text{sim}(I, T_j)/\tau)}$$

In the medical domain, this allows the model to learn that the visual pattern of “consolidation” is semantically equivalent to the textual description “consolidation in the lung,” enabling retrieval and classification without explicit labels. [4]

4.2 Zero-Shot Learning (ZSL) in Medicine

Standard deep learning requires thousands of labeled examples for every new disease class—a luxury rarely available in medicine due to the long tail of rare conditions. CLIP enables ZSL by allowing users to query the model with natural language prompts (e.g., “An X-ray of a patient with pneumonia”), effectively turning classification into a retrieval task.

4.2.1 Localization Mechanisms

- **Visual Prompting:** Techniques have been developed to query specific regions by leveraging the semantic alignment between text and image representations. For example, **Xplainer** [21] moves beyond simple class labels (e.g., “Pneumonia”) by using descriptive

prompts that break down a diagnosis into its constituent radiological signs (e.g., “opacities in the lung field” or “blunting of the costophrenic angle”). By querying the model with these granular descriptions, Xplainer explicitly guides the model’s attention to relevant anatomical regions, allowing for zero-shot localization that is both clinically accurate and inherently explainable.

- **Multimodal Concept Extraction:** Advanced methods extract fine-grained visual concepts from image patches and align them with textual concepts. *ConceptCLIP* enhances this by using a dual alignment strategy:
 - **IT-Align (Image-Text):** Matches the whole image to the full report to learn general context.
 - **PC-Align (Patch-Concept):** Forces the model to align specific image patches with specific medical concepts derived from the UMLS ontology. This dual objective allows the model to localize specific pathologies (e.g., a specific nodule) by matching image patches to concepts, even in a zero-shot setting, significantly outperforming standard CLIP which lacks this fine-grained alignment. [7]

4.3 Integrating Knowledge into VLMs

To overcome the “domain gap” between natural images (on which standard CLIP is trained) and medical images, and to reduce hallucinations, SOTA models explicitly integrate structured knowledge into the VLM architecture. A critical limitation of standard VLMs in medicine is their tendency to “hallucinate” findings—generating plausible but factually incorrect reports due to reliance on statistical correlations rather than causal medical logic [1]. For instance, a standard CLIP model might associate the word “pneumothorax” with “chest tube” due to frequent co-occurrence, leading it to hallucinate a tube in a healthy lung. Integrating Knowledge Graphs (KGs) serves as a necessary “safety guardrail” for these architectures. By grounding the model’s latent space in rigorous ontologies like RadLex or SNOMED CT, I impose physical and semantic constraints (e.g., *Pleural Effusion* must be located in the *Pleural Space*) that statistically-driven models lack. This structural grounding is essential for moving from high-accuracy benchmarks to safe, deployable clinical tools [6].

4.3.1 Subject-Level Enhancement

- **RadGraph:** This dataset and framework introduce a schema for extracting entities (Anatomy, Observation) and relations (Modify, Located At, Suggestive Of) from radiology reports. By training VLMs on RadGraph-XL (a large-scale dataset with over 410,000 expert annotations), models learn the specific syntactic and semantic structure of radiological reasoning. This structure acts as a strong prior, preventing the model from learning spurious associations. [6]

- **UMLS Integration:** Models like *MedKLIP* do not just use raw text; they utilize a *triplet extraction module* to convert reports into structured queries (e.g., {Observation: Effusion, Loc: Left Lower Lobe, Status: Present}) and map them to UMLS concepts. This effectively filters out linguistic noise and focuses the model on clinical facts. [5]

4.3.2 Domain-Level Enhancement

- **MedKLIP (Medical Knowledge Enhanced Language-Image Pre-training):** MedKLIP is a landmark framework that decouples raw report processing. It extracts medical entities from reports and queries an external knowledge base for detailed descriptions. It then uses a *transformer-based fusion module* to align these entity descriptions with image patches. Crucially, it employs a distinct loss function strategy: a *position contrastive loss (PosCL)* for localization and a cross-entropy loss for existence prediction. This decoupling allows MedKLIP to perform zero-shot grounding (localization) by querying the model with entity descriptions rather than simple class labels. [5]
- **MedFILIP:** A recent advancement that uses an LLM-based information extractor to decouple disease details and an “Image-specific Knowledge Injector” to construct relationships between categories and visual attributes. This facilitates better generalization to unseen diseases by leveraging attribute-level similarities. [22]
- **MeDSLIP:** This model introduces a dual-stream mechanism to explicitly disentangle pathology and anatomy semantics. It uses a *Prototypical Contrastive Loss (ProtoCL)* to align cross-modal information (e.g., pathology in image vs. anatomy in text) and an *Intra-image Contrastive Loss (ICL)* to model the co-existence of pathology and anatomy within the image. [23]
- **ItemizedCLIP:** This approach employs an “Itemized Local Alignment (ILA)” objective that aligns text items with corresponding visual evidence, enforcing inter-item separation to ensure distinct regions for distinct items, thereby improving fine-grained interpretability. [24]
- **MedBridge:** A lightweight adaptation framework that repurposes pretrained VLMs for medical diagnosis using a *Focal Sampling* module to extract high-resolution local regions, capturing subtle pathological features that general-purpose VLMs might miss due to resolution constraints. [25]
- **KEPIL:** A prompt-robust framework that integrates curated medical knowledge to stabilize zero-shot generalization. It uses dynamic prompt enrichment via ontologies and LLM assistance, and a semantic-aware contrastive loss to align embeddings of equivalent prompt variants. [26]

5 Evaluation Metrics

Evaluating S-XAI in medicine requires a departure from standard accuracy metrics (like AUC or Accuracy) toward metrics that assess explainability, localization fidelity, and clinical report quality.

5.1 Explainability & Concept Metrics

- **TCAV (Testing with Concept Activation Vectors):** Measures the sensitivity of a model’s prediction to a specific high-level concept. It quantifies the fraction of class inputs that are positively influenced by the concept vector, defined as the directional derivative of the logit with respect to the concept vector: [17]

$$TCAV_{Q,C,k} = \frac{|\{x \in X_k : \nabla h_{l,k}(f_l(x)) \cdot v_C^l > 0\}|}{|X_k|}$$

- **TCAR (Testing with Concept Activation Regions):** An evolution of TCAV for non-linear concepts. It measures the association between a class and a concept region in the latent space defined by a kernel SVC, providing a more accurate measure when concepts are not linearly separable. [18].
- **Faithfulness:** Measures how accurately the explanation reflects the true decision-making process of the model. Common metrics include: [27]
 - **Prediction Gap on Important feature perturbation (PGI):** Measures the change in prediction when top- k important features are perturbed.
 - **Prediction Gap on Unimportant feature perturbation (PGU):** Measures the change in prediction when unimportant features are perturbed.

5.2 Localization & Segmentation Metrics

- **Dice Score / Dice Loss:** Used to measure the overlap between the predicted attention map or segmentation mask and the ground truth region.
- **IoU (Intersection over Union):** The standard metric for object detection and zero-shot localization, calculated as the area of overlap divided by the area of union between the predicted and ground truth bounding boxes.
- **Pointing Game Accuracy:** Specifically used in MedKLIP and similar localization tasks. It evaluates whether the pixel with the maximum activation in the attention map falls within the ground truth bounding box of the pathology. [5]

5.3 Report Generation Quality

- **Diverse Sensitivity (DS):** Unlike BLEU, which measures n-gram overlap, DS specifically evaluates clinical accuracy. It checks whether the generated report correctly identifies the diseases present (sensitivity) and captures the diversity of valid disease concepts. It is particularly designed to penalize the “safe” generation of common diseases while ignoring rare conditions (the long-tail problem). [28]
- **Clinical Efficacy (CE) / CheXbert F1:** This metric uses a pre-trained labeler (like CheXbert) to extract clinical entities (diseases, observations) from the generated text and compares them to entities extracted from the ground truth report. It provides a measure of *clinical correctness* (Precision, Recall, F1) rather than linguistic similarity. [29]
- **NLP Metrics:** Standard metrics like BLEU, ROUGE, METEOR, and CIDEr are used for measuring textual fluency and similarity to reference reports. However, they are increasingly viewed as insufficient for medical AI because a report can be linguistically fluent but clinically disastrous (e.g., “No pneumothorax” vs. “Mild pneumothorax”). [11, 30]

5.4 Human-Centric and Trust Evaluation

While quantitative metrics like TCAR and Diverse Sensitivity (DS) measure technical correctness, they do not directly assess the clinical utility or “trustworthiness” of the system. In the context of S-XAI, “faithfulness” implies that the explanation actually assists the human expert [27]. Recent literature emphasizes the need for human-in-the-loop evaluation metrics:

- **Clinician Agreement Rate:** This measures the frequency with which a board-certified radiologist agrees with the model’s generated explanation (e.g., the specific region highlighted or the reasoning chain provided), rather than just the final classification label [12, 27].
- **Time-to-Diagnosis Reduction:** A practical metric evaluating whether the S-XAI explanations measurably reduce the time a radiologist spends reading a scan compared to a “black-box” baseline [2].
- **Intervention Efficacy:** As proposed in Concept Bottleneck Models, this measures how effectively a clinician can “correct” a model’s mistake by intervening on an intermediate concept (e.g., changing “irregular border” to “smooth”) and whether the model updates its final prediction accordingly [14, 18].

Although this thesis primarily focuses on computational frameworks, acknowledging these downstream clinical metrics is vital for contextualizing the proposed prototype’s potential real-world impact.

6 Comparative and Gap Analysis

6.1 Comparative Analysis of Approaches

Current approaches range from Post-hoc XAI (Grad-CAM), which suffers from low clinical alignment, to VLMs. Although standard CLIP offers zero-shot capability, it lacks the structured grounding of MedKLIP or the fine-grained concept alignment of ConceptCLIP.

Table 1: Comparison of Explainability Approaches in Medical Imaging

Feature	Post-hoc XAI (Grad-CAM, LIME)	Standard VLMs (CLIP)	Knowledge-Enhanced S-XAI {Proposed}
Explainability Source	Heuristic feature attribution (after training)	Attention maps (implicit)	Inherent architecture (CBMs, Triplet matching, KGs)
Clinical Alignment	Low; often highlights irrelevant regions or artifacts	Moderate; based on statistical text-image correlation	High; grounded in structured Ontologies (UMLS/RadLex)
Zero-Shot Capability	N/A (requires trained model)	High (Zero-shot classification)	High (Zero-shot + Concept discovery)
Transparency	Black Box	Black Box (mostly)	Glass Box (traceable reasoning)
Granularity	Coarse regions	Patch-level attention	Entity/Relation level (Fine-grained)

6.1.1 Comparison of Specific SOTA Models

- **MedKLIP vs. RadGraph:** RadGraph provides the data and schema (entities/relations) necessary for structured supervision, whereas MedKLIP is a model architecture that operationalizes this by filtering reports into triplets and using a query-based transformer for alignment. MedKLIP significantly outperforms standard methods in zero-shot settings because it relies on structured “medical queries” rather than noisy raw text [5,6].
- **Standard CLIP vs. ConceptCLIP:** Standard CLIP aligns global image and text vectors, often missing small, localized pathologies. ConceptCLIP introduces a PC-Align (Patch-Concept) branch, explicitly forcing the model to align specific image patches with medical concepts derived from UMLS. This results in superior interpretability and localization performance [18].
- **R2GenKG vs. Others:** R2GenKG distinguishes itself through the construction of a dynamic, hierarchical Multi-modal Knowledge Graph (M3KG) using GPT-4o. Unlike static

graph approaches, R2GenKG captures disease-aware visual tokens and supports multi-granularity reasoning, leading to superior report generation performance [11]. While R2GenKG effectively utilizes Hybrid KGs for generating text, its primary focus remains on the language decoder aspect of the VLM pipeline. It uses the graph to retrieve relevant textual patterns but does not explicitly constrain the *visual embedding space* for zero-shot localization of unseen pathologies. [11].

6.2 Gap Analysis (Identifying the “Missing Link”)

Despite significant progress, several critical gaps persist that limit the clinical utility of current S-XAI models:

1. **Disconnect between Zero-Shot & Explainability:** While CLIP provides robust zero-shot classification, it remains a “black box” regarding why it localized a specific region. It lacks an explicit reasoning chain. Current ZSL methods often rely on attention heatmaps that are not causally linked to specific clinical criteria, meaning the model might look at the right place for the wrong reason (e.g., a chest tube rather than the pneumothorax itself). [1]
2. **Lack of Structured Grounding:** Most VLMs rely on statistical co-occurrences in noisy web text or reports. They lack a “world model” or structured medical logic. While Med-KLIP and ConceptCLIP start to address this with UMLS, they often use static definitions. There is a need for dynamic, context-aware grounding that can handle the “long tail” of rare diseases better. [28]
3. **Scalability of Concept Models:** Supervised Concept Bottleneck Models (CBMs) require expensive, dense annotations for every concept, which is unscalable. Unsupervised methods (ACE) often discover incoherent concepts. CEIR bridges this with generative concepts from GPT-4, but ensuring these generated concepts are clinically faithful and not just semantically plausible remains a challenge. [19]

6.3 How this Thesis Fills the Gap

This thesis proposes a framework embedding **Hybrid Knowledge Graphs** directly into **CLIP training**. By integrating expert ontologies (RadLex) with data-driven insights (GPT-4o), the proposed method bridges the gap between “unseen disease detection” (Zero-shot) and “clinical justification” (S-XAI). Specifically, it targets the creation of a system where zero-shot localization is not merely an artifact of attention, but the result of a traceable query against a structured medical knowledge base. This will be rigorously evaluated using clinical metrics like Diverse Sensitivity (DS) and faithfulness measures (TCAR), moving beyond simple linguistic overlap.

7 Contribution of Literature Review to Thesis

This literature review serves as the theoretical bedrock for the thesis, directly informing the design choices and evaluation strategies of the proposed S-XAI framework.

7.1 Self-Explainable AI (S-XAI) for Medical Imaging

The review establishes that S-XAI is the necessary evolution for medical AI. By detailing architectures like CBMs and Attention-Branch Networks, the review justifies the thesis’s choice to abandon post-hoc methods in favor of inherently interpretable designs. The detailed analysis of CEIR provides a specific roadmap for utilizing Generative AI (GPT-4) to create concept bottlenecks, a strategy the thesis will adapt and refine for the medical domain. [19]

7.2 Contrastive Language-Image Pretraining (CLIP) for Zero-Shot Localization

The deep technical dive into CLIP, MedKLIP, and ConceptCLIP validates the use of contrastive learning as the primary mechanism for handling data scarcity. The review identifies PC-Align (Patch-Concept Alignment) as a critical mechanism for achieving fine-grained localization. The thesis will build upon this by integrating dynamic graph-based queries into the alignment process, rather than static concept lists. [5, 7]

7.3 Knowledge Graphs and Medical Ontologies for Enhancing AI Transparency

The analysis of RadGraph and R2GenKG demonstrates that structured knowledge is essential for reducing hallucinations and ensuring clinical accuracy. The distinction between Prior, Data-Driven, and Hybrid KGs informs the thesis’s approach to constructing a Hybrid KG that leverages the strengths of both expert rules (RadLex) and LLM-generated insights (M3KG), aiming to solve the long-tail distribution problem. [6, 11]

7.4 Explainable Medical Report Generation

Finally, the review of report generation metrics—specifically Diverse Sensitivity (DS) and Clinical Efficacy (CE)—provides the evaluation framework for the thesis. It highlights that while generating “fluent” text is largely solved by modern LLMs, the remaining challenge is generating clinically accurate and diverse reports for rare conditions. The thesis aims to solve this through KG-enhanced zero-shot learning, and these metrics will be the primary yardstick for success. [28]

8 Conclusion

The transition from black-box deep learning to Self-Explainable AI is not merely a technical upgrade but an ethical and clinical necessity in medical imaging. This literature review has synthesized the rapid advancements in this field, highlighting the transformative role of CLIP-based Vision-Language Models and the grounding power of Knowledge Graphs. While models like MedKLIP, ConceptCLIP, and R2GenKG have set new benchmarks in zero-shot localization and report generation, significant gaps remain in unifying these capabilities into a cohesive, clinically transparent framework. By integrating Hybrid Knowledge Graphs directly into the CLIP pre-training and fine-tuning process, and by rigorously evaluating performance using clinical metrics like Diverse Sensitivity, the proposed thesis stands poised to make a substantial contribution to the development of trustworthy, transparent, and interpretable medical AI. The path forward lies in systems that do not just "see" patterns, but "understand" medical concepts, fostering a collaborative partnership between AI and clinicians.

9 Methodology

This section details the proposed framework for a Knowledge-Infused Self-Explainable AI (SXAI) system designed for medical imaging. I present a neuro-symbolic architecture that integrates a Hybrid Knowledge Graph with a Vision-Language Model (VLM) to achieve both zero-shot localization and faithful clinical report generation. The approach moves beyond purely data-driven correlations by grounding visual representations in structured medical ontologies.

9.1 Mathematical Problem Formulation

I formulate the task as a multi-modal mapping problem. Let $\mathcal{D} = \{(x_i, r_i)\}_{i=1}^N$ denote a dataset of N pairs, where $x_i \in \mathbb{R}^{H \times W \times C}$ represents a medical image (e.g., chest X-ray) and r_i represents the corresponding radiology report. The report is defined as a sequence of tokens $r_i = \{t_1, t_2, \dots, t_L\}$.

I further define a **Knowledge Graph** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is a set of medical entities derived from established ontologies (RadLex/UMLS) and $\mathcal{E}$ represents the semantic relationships between them (e.g., *Located-At*, *Is-A*).

The core objective is to learn a parameterized mapping function $f_\theta : x \rightarrow (\hat{r}, \mathcal{A})$ that outputs:

1. A generated clinical report $\hat{r}$ that is clinically faithful to the image content.
2. A set of **Concept Activation Maps** $\mathcal{A} = \{M_c\}_{c \in \mathcal{C}}$, where each map $M_c \in \mathbb{R}^{H \times W}$ explicitly highlights the visual evidence for a specific medical concept c , thereby providing inherent explainability.

9.2 Phase I: Construction of the Hybrid Knowledge Graph

To overcome the limitations of static knowledge bases, which may lack dataset-specific nuances, and purely data-driven approaches, which are prone to hallucinations, I construct a **Hybrid Knowledge Graph**. This design follows the principles of recent neuro-symbolic frameworks such as R2GenKG [11] and MedKLIP [5], enabling the integration of structured medical knowledge with dataset-specific visual evidence.

9.2.1 Ontology-Guided Entity Extraction

I employ a deterministic entity extraction and normalization pipeline grounded in the **RadGraph-XL** dataset [13]. Structured entities and relations are extracted using the RadGraph inference engine based on DYGIE++ [6], yielding a set of instance-level triplets $(e_{head}, rel, e_{tail})$ directly derived from radiology reports.

The extraction process is constrained as follows:

- **Entity Types:** Restricted to ANATOMY and OBSERVATION. Numeric MEASUREMENT entities are excluded from ontology grounding, as they do not correspond to stable biomedical concepts.
- **Relation Types:** All RadGraph-predicted relations (e.g., LOCATED_AT, MODIFY, SUGGESTIVE_OF) are preserved without introducing additional relational assumptions.

Each extracted entity mention is subsequently mapped to a **UMLS Concept Unique Identifier (CUI)** using a deterministic, exact-match normalization procedure based on the UMLS MRCONSO vocabulary. This step resolves lexical variation and synonymy across reports (e.g., mapping surface variants of radiologic findings to standardized ontology identifiers) without relying on heuristic confidence thresholds or fuzzy matching.

9.2.2 Hybrid Graph Structure

The resulting graph $\mathcal{G}$ integrates instance-level observations with ontology-anchored medical concepts and comprises two complementary node categories:

- **Instance Nodes ($\mathcal{V}_{data}$):** Report-specific entities extracted from the training batch (e.g., “Left Lower Lobe Opacity”), connected via relations predicted by the RadGraph inference engine.
- **Ontology-Anchored Concept Nodes ($\mathcal{V}_{prior}$):** UMLS concepts identified via CUI normalization, which provide a shared semantic reference space across reports and enable aggregation of rare or lexically diverse findings.

By anchoring instance-level entities to standardized ontology concepts while preserving data-driven relational structure, the hybrid graph mitigates the long-tail distribution commonly

observed in medical datasets [28]. Rare conditions benefit from shared ontology-level representations even when direct training examples are limited, without introducing externally imposed relational bias.

9.3 Phase II: The Neuro-Symbolic Architecture

I propose a Tri-Stream Architecture comprising a Visual Encoder, a Knowledge Encoder, and a specialized Fusion Module. This design is chosen to explicitly facilitate the interaction between pixel-level data and symbolic knowledge.

9.3.1 Visual Encoder (E_V)

I adopt a **Vision Transformer (ViT-B/16)** as the visual backbone. I initialize the model with **BioMedCLIP** weights [1], rather than standard ImageNet weights.

$$Z_v = E_V(x) \in \mathbb{R}^{P \times D_v} \quad (1)$$

where P denotes the number of patches and D_v is the visual embedding dimension.

Justification: ViT architectures process images as sequences of patches, which aligns conceptually with the discrete nature of knowledge graph nodes, facilitating more effective cross-modal attention. Furthermore, BioMedCLIP initialization provides domain-specific feature representations that are superior for medical tasks compared to general-domain pre-training [1].

9.3.2 Knowledge Encoder (E_K)

To process the structured information from $\mathcal{G}$, I employ a **Graph Attention Network (GAT)**. Unlike simple Graph Convolutional Networks (GCNs), GATs learn dynamic attention weights for different edge types, allowing the model to prioritize semantically relevant relationships (e.g., emphasizing *Located-At* edges for localization tasks) [11].

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} W h_j^{(l)} \right) \quad (2)$$

where α_{ij} represents the learned attention weight between medical concepts. The output is a set of refined concept embeddings $Z_k \in \mathbb{R}^{K \times D_k}$.

9.3.3 Knowledge-Visual Fusion via Cross-Attention

To achieve grounded concept identification, I implement a **Query-Based Fusion** mechanism, inspired by architectures such as MedKLIP [5] and ConceptCLIP [7]. I explicitly reject simple concatenation in favor of an attention-based bottleneck.

I treat the Knowledge Concepts (Z_k) as **Queries** (Q) and the Visual Patches (Z_v) as **Keys** (K) and **Values** (V):

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{Z_k W_Q (Z_v W_K)^T}{\sqrt{d}} \right) (Z_v W_V) \quad (3)$$

Result: This operation generates *Concept-Aware Visual Features*. Effectively, the model uses medical concepts to "query" the image, forcing the attention mechanism to highlight regions relevant to specific pathologies. This directly enables the extraction of concept activation maps for explainability [2].

9.4 Phase III: Disentangled Decoder for Report Generation

To ensure methodological rigor, I enforce a strict separation between representation learning and text generation.

9.4.1 Decoder Specification (D_{gen})

I utilize a Transformer Decoder, specifically a lightweight **DistilGPT** or specialized **RadGPT**, for the generation task.

- **Conditioning:** The decoder is conditioned *solely* on the fused Concept-Aware Visual Features derived in Phase II.
- **Mechanism:** The generation probability is modeled as $P(t_i | t_{<i}, Z_{fused})$.

Justification: By adhering to the standard Encoder-Decoder paradigm and isolating the decoder from raw pixel data, I constrain the generator to rely on the detected medical concepts. This structure serves as a regularization mechanism to reduce hallucinations, ensuring the output is grounded in the identified clinical entities [30].

9.5 Phase IV: Training Objectives

I employ a multi-task objective function ($\mathcal{L}_{total}$) designed to optimize for both zero-shot alignment and generation quality.

9.5.1 Contrastive Alignment Loss ($\mathcal{L}_{align}$)

To enable zero-shot capability, I maximize the mutual information between visual embeddings and clinical concepts using the InfoNCE loss [4]:

$$\mathcal{L}_{align} = -\log \frac{\exp(\text{sim}(z_v, z_k)/\tau)}{\sum_j \exp(\text{sim}(z_v, z_{k_j})/\tau)} \quad (4)$$

This aligns image patches with their corresponding graph nodes in the latent space.

9.5.2 Entity Existence Loss ($\mathcal{L}_{exist}$)

I apply a binary cross-entropy loss to the output of the fusion layer to classify the presence or absence of specific medical concepts:

$$\mathcal{L}_{exist} = -\frac{1}{N} \sum_{i=1}^N y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i) \quad (5)$$

This objective forces the Knowledge Encoder to function as a multi-label classifier, ensuring that the concepts passed to the generator are factually present in the image [25].

9.5.3 Language Modeling Loss ($\mathcal{L}_{LM}$)

I utilize the standard autoregressive Cross-Entropy loss for the report generation decoder.

The final objective is a weighted sum:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{align} + \lambda_2 \mathcal{L}_{exist} + \lambda_3 \mathcal{L}_{LM} \quad (6)$$

9.6 Experimental Setup and Evaluation Strategy

9.6.1 Datasets and Tools

The model is trained on the **MIMIC-CXR** dataset, utilizing the **RadGraph-XL** annotations for graph construction. Implementation is performed using **PyTorch**, with the **RadGraph** inference engine used for preprocessing.

9.6.2 Evaluation Metrics

I implement a three-tier evaluation protocol to assess Explainability, Accuracy, and Faithfulness.

Tier 1: Zero-Shot Localization (Explainability)

- **TCAR (Testing with Concept Activation Regions):** I measure the alignment of activation regions in the latent space with positive concept examples to assess concept separability [18].
- **Pointing Game Accuracy:** I evaluate localization precision by checking if the maximum pixel in the attention map for a concept (e.g., "Pneumonia") falls within the ground-truth bounding box provided by the **MS-CXR** or **RSNA Pneumonia** benchmarks [5].

Tier 2: Report Generation (Clinical Accuracy)

- **Clinical Efficacy (CE) / RadGraph F1:** Following standard practice [29], I prioritize CE over n-gram metrics like BLEU. I use the CheXbert labeler to extract entities from both generated and ground-truth reports and compute the F1 score of their overlap.

- **Diverse Sensitivity (DS):** To address the long-tail problem, this metric penalizes the model for ignoring rare diseases and over-predicting common ones [28].

Tier 3: Faithfulness (Trust)

- **Intervention Efficacy:** Inspired by Concept Bottleneck Models [14], I perform intervention experiments where I manually suppress a concept vector (e.g., setting "Pleural Effusion" to zero) in the fusion layer. A faithful model must subsequently cease mentioning the finding in the generated report, demonstrating a causal link between the internal representation and the output.

9.7 Baselines and Ablation

To validate the proposed framework, I compare performance against:

1. **CNN-RNN (ResNet-50 + LSTM):** A traditional baseline.
2. **Standard VLM (CLIP + GPT-2):** Represents performance without structured knowledge infusion.
3. **MedKLIP:** The current state-of-the-art competitor [5].

An ablation study will further isolate the contribution of the Hybrid Graph by comparing models trained with No Graph, a Static (UMLS-only) Graph, and the proposed Hybrid Graph.

References

- [1] Z. Zhao, Y. Liu, H. Wu *et al.*, "Clip in medical imaging: A comprehensive survey," *Medical Image Analysis*, vol. 102, p. 103551, 2025.
- [2] J. Hou, S. Liu, Y. Bie *et al.*, "Self-explainable ai for medical image analysis: A survey and new outlooks," *arXiv preprint arXiv:2410.02331*, 2024.
- [3] L. Li *et al.*, "Scouter: Slot attention-based classifier for explainable image recognition," in *ICCV*, 2021.
- [4] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *ICML*, 2021.
- [5] C. Wu, X. Zhang, Y. Zhang, Y. Wang, and W. Xie, "Medklip: Medical knowledge enhanced language-image pre-training for x-ray diagnosis," *ICCV*, 2023.
- [6] S. Jain, A. Agrawal, A. Saporta *et al.*, "Radgraph: Extracting clinical entities and relations from radiology reports," in *NeurIPS Datasets and Benchmarks Track*, 2021.

- [7] Y. Nie, S. He, Y. Bie *et al.*, “Conceptclip: An explainable biomedical foundation model via large-scale concept-enhanced vision-language pre-training,” *arXiv preprint arXiv:2501.15579*, 2025.
- [8] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” *NeurIPS*, 2017.
- [9] S. Kapse *et al.*, “Si-mil: Taming deep mil for self-interpretability in gigapixel histopathology,” in *CVPR*, 2024.
- [10] I. Santos and A. de Carvalho, “Protoal: Interpretable deep active learning with prototypes for medical imaging,” *arXiv preprint arXiv:2404.04736*, 2024.
- [11] F. Wang, Y. Qiao, X. Wang *et al.*, “R2genkg: Hierarchical multi-modal knowledge graph for llm-based radiology report generation,” in *arXiv preprint arXiv:2508.03426*, 2025.
- [12] C. Goh, J. Ng, W. Y. Au, C. See, A. Lim, J. W. Zheng, X. Fan, and K. Li, “Evaluating a multi-modal large language model for ophthalmology triage,” *Journal of Clinical & Translational Ophthalmology*, 2025.
- [13] J.-B. Delbrouck, P. Chambon, Z. Chen *et al.*, “Radgraph-xl: A large-scale expert-annotated dataset for entity and relation extraction from radiology reports,” in *Findings of ACL*, 2024.
- [14] P. W. Koh, T. Nguyen, Y. S. Chi *et al.*, “Concept bottleneck models,” in *ICML*, 2020.
- [15] H. Wang, J. Hou, S. He *et al.*, “Concept complement bottleneck model for interpretable medical image diagnosis,” *MIDL Conference Submission*, 2026.
- [16] T. Oikarinen, S. Das, L. M. Nguyen, and T.-W. Weng, “Label-free concept bottleneck models,” in *International Conference on Learning Representations*, 2023.
- [17] B. Kim, M. Wattenberg, J. Gilmer *et al.*, “Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav),” in *ICML*, 2018.
- [18] J. Crabbé and M. van der Schaar, “Concept activation regions: A generalized framework for concept-based explanations,” in *NeurIPS*, 2022.
- [19] Y. Cui, S. Liu, L. Li, and Z. Yuan, “Ceir: Concept-based explainable image representation learning,” *arXiv preprint arXiv:2312.10747*, 2023.
- [20] M. Dikter, T. Blau, and C. Baskin, “Conceptual learning via embedding approximations for reinforcing interpretability and transparency,” in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025.

- [21] C. Pellegrini, M. Keicher, E. Özsoy, P. Jiraskova, R. Braren, and N. Navab, “Xplainer: From x-ray observations to explainable zero-shot diagnosis,” *arXiv preprint arXiv:2303.13391*, 2023.
- [22] X. Liang, X. Li, F. Li, J. Jiang, Q. Dong, W. Wang, K. Wang, S. Dong, G. Luo, and S. Li, “Medfilip: Medical fine-grained language-image pre-training,” *arXiv preprint arXiv:2501.10775*, 2025.
- [23] Y. Luo, Y. Zhang, and V. Osmani, “Medslip: Medical dual-stream language-image pre-training for pathology-anatomy alignment,” in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. Springer, 2024, pp. 35–45.
- [24] Y. Luo, Y. Zhang, and X. Zhang, “Learning complete and explainable visual representations from itemized text supervision,” *arXiv preprint arXiv:2501.10775*, 2025.
- [25] Y. Li, M. Ghahremani, and C. Wachinger, “Medbridge: Bridging foundation vision-language models to medical image diagnosis,” *arXiv preprint arXiv:2505.21698*, 2025.
- [26] Anonymous, “Kepil: Knowledge-enhanced prompt-image learning for prompt-robust disease detection,” OpenReview, 2025.
- [27] A. Pratinav Seth *et al.*, “Bridging the gap in xai—the need for reliable metrics in explainability and compliance,” *arXiv preprint arXiv:2502.04695*, 2025.
- [28] Y. Liu, X. Li, J. Tang, and B. Zhang, “Rethinking medical report generation: Disease revealing enhancement with knowledge graph,” *arXiv preprint arXiv:2106.06968*, 2021.
- [29] P. Wang, S. Ye, U. Naseem, and J. Kim, “Mrg-r1: Reinforcement learning for clinically aligned medical report generation,” *arXiv preprint arXiv:2512.16145*, 2025.
- [30] Ș.-V. Voinea, M. Mămuleanu, R. V. Teică, L. M. Florescu, D. Selișteanu, and I. A. Gheonea, “Gpt-driven radiology report generation with fine-tuned llama 3,” *Bioengineering*, 2024.